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COURSE: BACHELORS IN COMPUTER SCIENCE

UNIT NAME: ETHICAL HACKING

UNIT CODE: BIT 4139

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SEMESTER: YEAR 3 SEM 2

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COMPLETE INSTALLATION

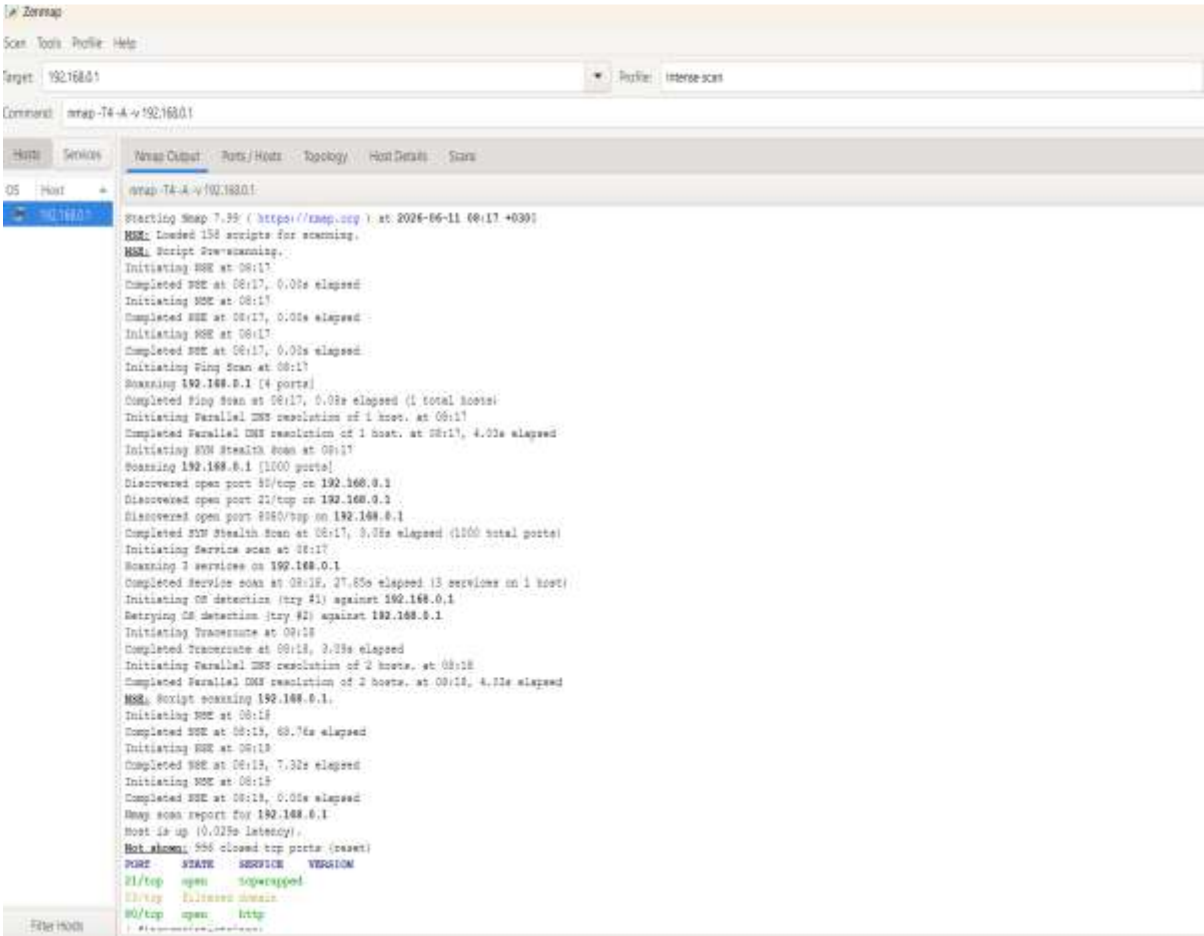


Figure 1: Nmap scan of router (192.168.0.1) showing open HTTP service and device detail

FOOTPRINTING

Host Discovery

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\user> nmap -sn 192.168.1.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-28 08:15 +0300
Nmap scan report for 192.168.1.1
Host is up (0.011s latency).
Nmap scan report for 192.168.1.20
Host is up (0.11s latency).
Nmap scan report for 192.168.1.21
Host is up (0.11s latency).
Nmap scan report for 192.168.1.44
Host is up (0.040s latency).
Nmap scan report for 192.168.1.85
Host is up (0.038s latency).
Nmap scan report for 192.168.1.107
Host is up (0.050s latency).
Nmap scan report for 192.168.1.175
Host is up (0.21s latency).
Nmap scan report for 192.168.1.186
Host is up (0.010s latency).
Nmap scan report for 192.168.1.209
Host is up (0.019s latency).
Nmap scan report for 192.168.1.236
Host is up (0.029s latency).
Nmap scan report for 192.168.1.237
Host is up (0.028s latency).
Nmap scan report for 192.168.1.243
Host is up (0.023s latency).
Nmap scan report for 192.168.1.244
Host is up (0.023s latency).
Nmap scan report for 192.168.1.250
Host is up (0.026s latency).
```

Figure 2: Nmap host discovery scan showing active devices on the local network (192.168.1.0/24)

```
   Link-local IPv6 Address . . . . . : fe80::c33f:3d4c:c42d:2d0c%20
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter WiFi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::992a:9000:2c24:c88a%21
   IPv4 Address. . . . . . . . . . . : 192.168.0.102
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.0.1

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter vEthernet (Default Switch):

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::7744:bc8a:c5ab:b586%22
   IPv4 Address. . . . . . . . . . . : 172.24.160.1
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . :
```

Figure 3: Windows IP configuration showing local network details and active network interfaces

SCANNING NETWORKS

Port Scanning

```
PS C:\Users\user> nmap 192.168.1.5
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-28 08:25 +0300
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.20 seconds
PS C:\Users\user> nmap 192.168.0.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-28 08:25 +0300
Nmap scan report for 192.168.0.1
Host is up (0.0067s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE
80/tcp open  http
MAC Address: CC:2D:21:A0:18:D0 (Tenda Technology,Ltd.Dongguan branch)

Nmap done: 1 IP address (1 host up) scanned in 1.91 seconds
PS C:\Users\user>
```

Figure 4: Nmap host discovery scan executed in Windows PowerShell showing active devices within the local subnet and initial network reconnaissance results

```
Nmap done: 1 IP address (1 host up) scanned in 1.91 seconds
PS C:\Users\user> nmap -sV 192.168.1.5
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-28 08:28 +0300
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.31 seconds
PS C:\Users\user> nmap -sV 192.168.0.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-28 08:28 +0300
Nmap scan report for 192.168.0.1
Host is up (0.0075s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
80/tcp open  http    GoAhead WebServer
MAC Address: CC:2D:21:A0:18:D0 (Tenda Technology,Ltd.Dongguan branch)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.81 seconds
PS C:\Users\user>
```

Figure 5: Nmap service version detection scan in Windows PowerShell identifying the specific web server application running on port 80 of the active target.

VULNERABILITY SCANNING

Tool: Nmap

```
PS C:\Users\user> nmap -A 192.168.1.5
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-28 08:30 +0300
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.44 seconds
PS C:\Users\user> nmap -A 192.168.0.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-28 08:30 +0300
Nmap scan report for 192.168.0.1
Host is up (0.0035s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
80/tcp open  http    GoAhead WebServer
| http-title: Tenda Wireless Router
|_Requested resource was http://192.168.0.1/index.html
MAC Address: CC:2D:21:A0:18:D0 (Tenda Technology,Ltd.Dongguan branch)
Device type: general purpose
Running: Wind River VxWorks
OS CPE: cpe:/o:windriver:vxworks
OS details: VxWorks
Network Distance: 1 hop

TRACEROUTE
HOP RTT     ADDRESS
1   3.49 ms 192.168.0.1

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 20.42 seconds
PS C:\Users\user>
```

Figure 6: Nmap aggressive scan (-A) output showing operating system detection and HTTP script results for the target host.

```
PS C:\Users\user> nmap --script vuln 192.168.1.5
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-28 08:32 +0300
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 15.98 seconds
PS C:\Users\user> nmap --script vuln 192.168.0.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-28 08:33 +0300
Nmap scan report for 192.168.0.1
Host is up (0.0052s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE
80/tcp open  http
|_http-dombased-xss: Couldn't find any DOM based XSS.
|_http-csrf: Couldn't find any CSRF vulnerabilities.
| http-fileupload-exploiter:
|
|_    Couldn't find a file-type field.
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
MAC Address: CC:2D:21:A0:18:D0 (Tenda Technology,Ltd.Dongguan branch)

Nmap done: 1 IP address (1 host up) scanned in 235.83 seconds
PS C:\Users\user>
```

Figure 7: Nmap vulnerability scan (--script vuln) output displaying automated security assessment results for the open HTTP service on the target host.

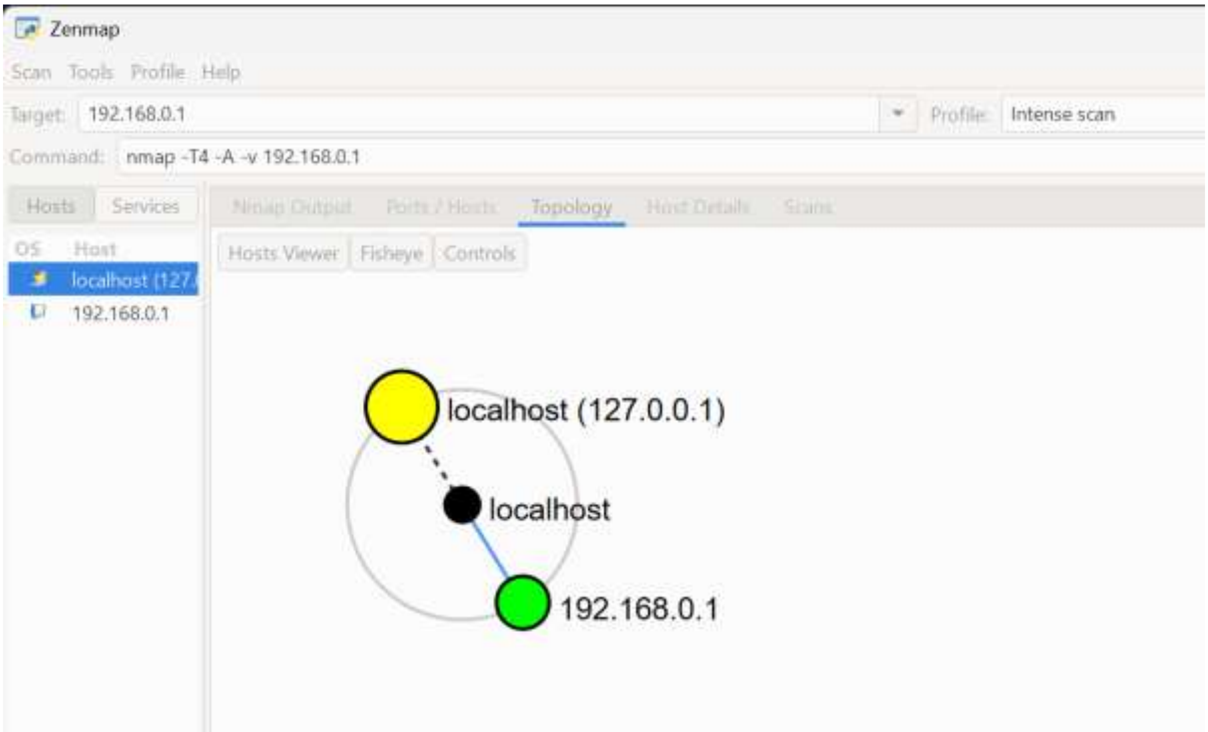


Figure 8: Zenmap graphical user interface displaying the network topology and node relationship between the local host and the target IP address.

Summary

INFORMATION GATHERING

SAMPLE TASK 1

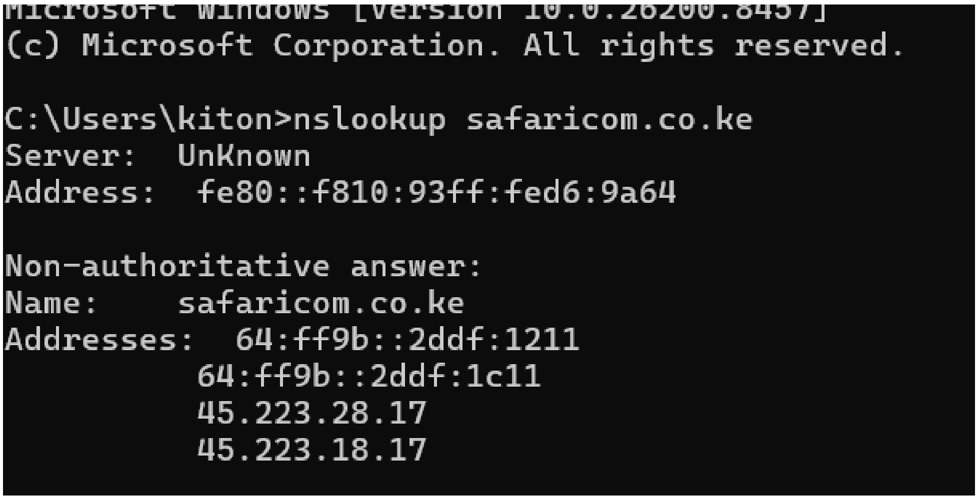
Company Background

Safaricom

Safaricom : Premier Mobile, Data, & M-PESA Services

Tools used

- WHOIS
- NSLOOKUP
- Wappalyzer
- crt.sh
- Have I Been Pwned
- Maltego


```
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\kiton>nslookup safaricom.co.ke
Server:  UnKnown
Address:  fe80::f810:93ff:fed6:9a64

Non-authoritative answer:
Name:    safaricom.co.ke
Addresses:  64:ff9b::2ddf:1211
          64:ff9b::2ddf:1c11
          45.223.28.17
          45.223.18.17
```

Figure 9: Nslookup results showing DNS resolution and IP address information for the KRA domain (kra.go.ke).

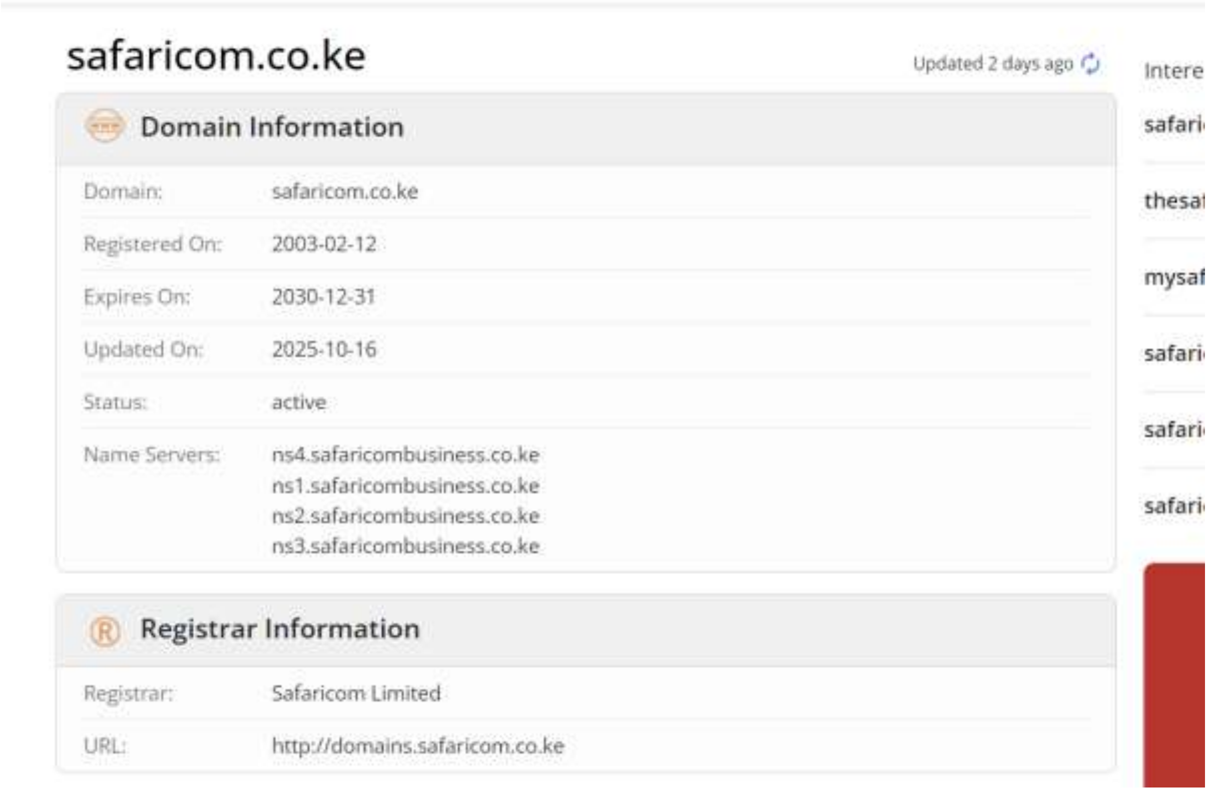


Figure 10: Domain Information

Registrar Information

Registrar:	Safaricom Limited
URL:	http://domains.safaricom.co.ke

Raw Whois Data

```
Domain Name: safaricom.co.ke
Registry Domain ID: 1403-Kenic
Registrar URL: http://domains.safaricom.co.ke
Updated Date: 2025-10-16T06:36:04Z
Creation Date: 2003-02-12T21:00:00Z
Registry Expiry Date: 2030-12-31T21:00:00Z
Registrar Registration Expiration Date: 2030-12-31T21:00:00Z
Registrar: Safaricom Limited
Registrar Street Address: Safaricom House, P.O. Box 46350 - 001001, Nairobi
Registrar Phone: 0722002222
Registrar Email: ebt-domains@Safaricom.co.ke
Registrar Abuse Contact Email:
Registrar Abuse Contact Phone:
Domain Status: active https://icann.org/epp#active
Name Server: ns4.safaricombusiness.co.ke
Name Server: ns1.safaricombusiness.co.ke
Name Server: ns2.safaricombusiness.co.ke
Name Server: ns3.safaricombusiness.co.ke
DNSSEC: signedDelegation
>>> Last update of WHOIS database: 2026-06-08T10:08:33.756Z <<<
```

Figure 11: Registrar Information

```
Default Server:  UnKnown
Address:  fe80::f810:93ff:fed6:9a64

> set type=ns
> safaricom.co.ke
Server:  UnKnown
Address:  fe80::f810:93ff:fed6:9a64

Non-authoritative answer:
safaricom.co.ke nameserver = ns1.safaricombusiness.co.ke
safaricom.co.ke nameserver = ns2.safaricombusiness.co.ke
safaricom.co.ke nameserver = ns3.safaricombusiness.co.ke
safaricom.co.ke nameserver = ns4.safaricombusiness.co.ke
> nmap -sV safaricom.co.ke
Unrecognized command: nmap -sV safaricom.co.ke
>
C:\Users\kiton>nmap -sV safaricom.co.ke
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-11 12:05 +0300
```

Figure 12: Open ports scan and services

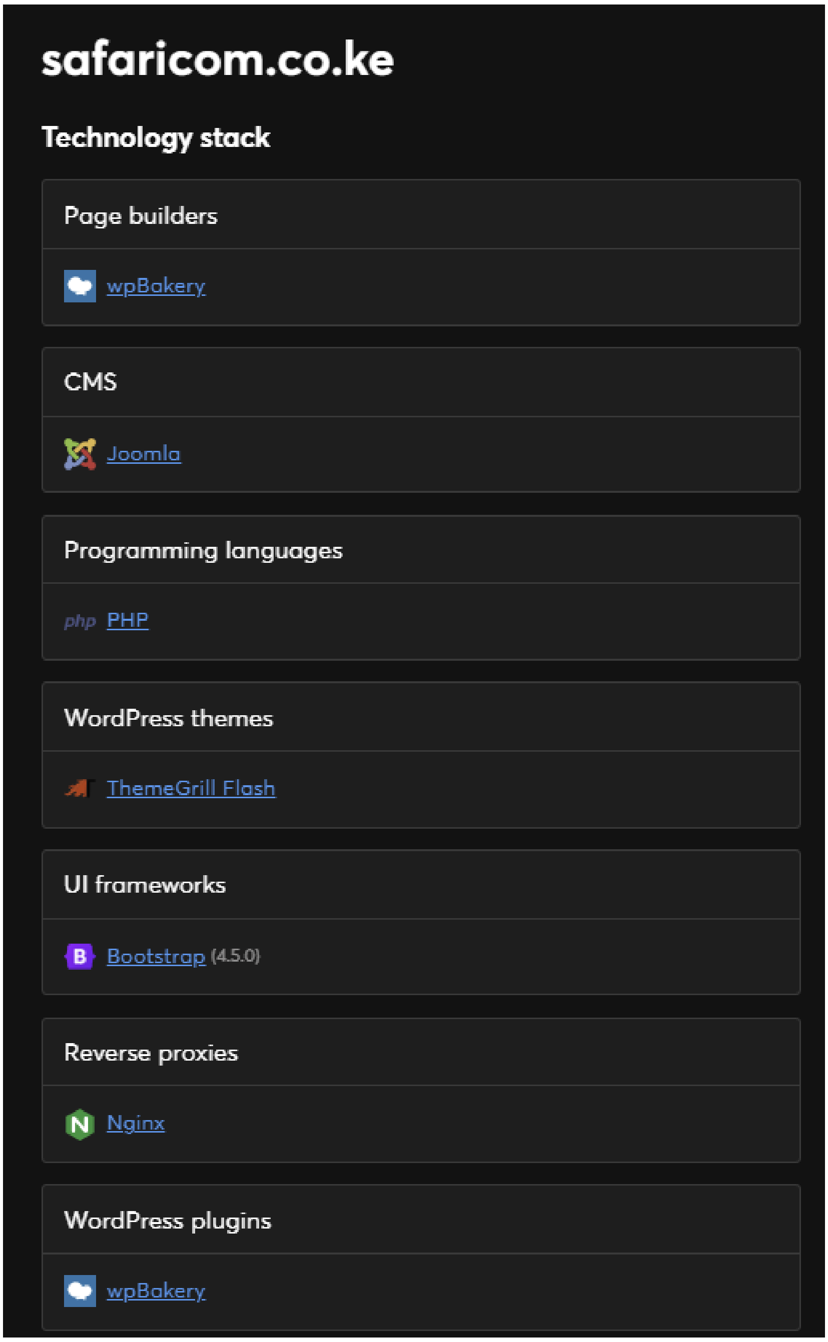


Figure13: Wappalyzer showing technologies used

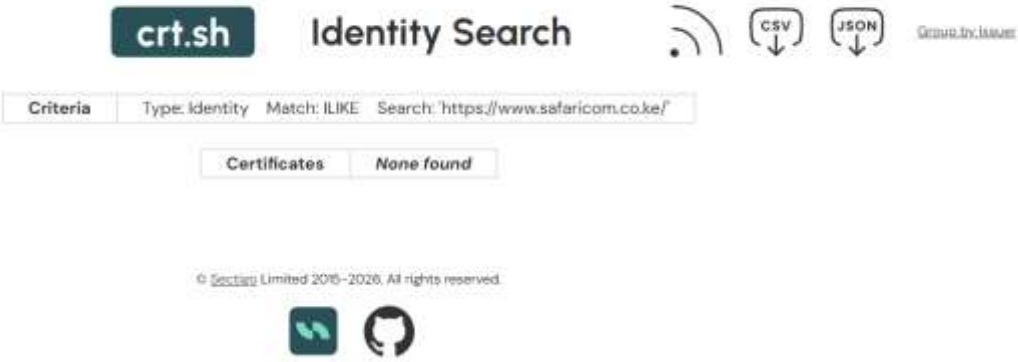


Figure 14: Results from crt.sh showing no Certificate Transparency records

Figure 15: Intelligence X search results for https://www.safaricom.co.ke/, showing publicly indexed files and documents associated with the domain.

Figure 16: Have I Been Pwned results for mobileoffice@safaricom.co.ke confirming the email address has appeared in 7 data breaches, indicating significant credential exposure risk.

```
C:\Users\kiton>nmap -sV safaricom.co.ke
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-11 14:33 +0300
^C
C:\Users\kiton>nmap -p 80,443 safaricom.co.ke
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-11 14:48 +0300
Nmap scan report for safaricom.co.ke (45.223.28.17)
Host is up (0.082s latency).
Other addresses for safaricom.co.ke (not scanned): 45.223.18.17 64:ff9b::2ddf:1211 64:ff9b::2ddf:1c11

PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 4.96 seconds

C:\Users\kiton>
```

Figure 17: Nmap ACK scan on Safaricom.co.ke showing all scanned ports in a filtered state with no response, confirming the presence of a firewall protecting the organization's infrastructure.

Task 2: Scan Specific Machine

```
   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::56e5:191f:dbbb:9a62%19
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter WiFi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2c0f:fe38:2215:49e2:e668:ce2b:a202:8e8
   Temporary IPv6 Address. . . . . . : 2c0f:fe38:2215:49e2:ddc6:151f:c5ef:9c44
   Link-local IPv6 Address . . . . . : fe80::992a:9000:2c24:c88a%20
   IPv4 Address. . . . . . . . . . . : 10.49.135.34
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::c088:47ff:febe:b5ac%20
                                       10.49.135.199

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter vEthernet (Default Switch):

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::f308:fc21:5e83:b25%39
   IPv4 Address. . . . . . . . . . . : 172.21.16.1
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . :

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::1d4:47ed:4720:5713%63
   IPv4 Address. . . . . . . . . . . : 172.19.96.1
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
```

Figure18: Ipconfig output displaying network adapter configurations and IP addresses of the host machine.

```
C:\Users\user>arp -anmap -p- 192.168.1.0

Displays and modifies the IP-to-Physical address translation tables used by
address resolution protocol (ARP).

ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]

  -a            Displays current ARP entries by interrogating the current
                protocol data.  If inet_addr is specified, the IP and Physical
                addresses for only the specified computer are displayed.  If
                more than one network interface uses ARP, entries for each ARP
                table are displayed.
  -g            Same as -a.
  -v            Displays current ARP entries in verbose mode.  All invalid
                entries and entries on the loop-back interface will be shown.
  inet_addr     Specifies an internet address.
  -N if_addr    Displays the ARP entries for the network interface specified
                by if_addr.
  -d            Deletes the host specified by inet_addr. inet_addr may be
                wildcarded with * to delete all hosts.
  -s            Adds the host and associates the Internet address inet_addr
                with the Physical address eth_addr.  The Physical address is
                given as 6 hexadecimal bytes separated by hyphens. The entry
                is permanent.
  eth_addr      Specifies a physical address.
  if_addr       If present, this specifies the Internet address of the
                interface whose address translation table should be modified.
                If not present, the first applicable interface will be used.
Example:
  > arp -s 157.55.85.212   00-aa-00-62-c6-09  .... Adds a static entry.
  > arp -a                                    .... Displays the arp table.
```

Figure19: Scan Ports, ARP command help page displayed due to incorrect syntax entry.

```
C:\Users\user>nmap -sV 192.168.1.0
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-04 00:31 +0300
Nmap scan report for 192.168.1.0
Host is up (0.023s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE    SERVICE VERSION
53/tcp filtered domain

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 24.39 seconds
```

Figure 20: Nmap service version scan on 192.168.1.0 revealing one open port — port 53 (DNS) in a filtered state, with 999 closed TCP ports detected.

```
C:\Users\user>nmap -O 192.168.1.0
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-04 00:34 +0300
Nmap scan report for 192.168.1.0
Host is up (0.024s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE    SERVICE
53/tcp filtered domain
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: WAP|printer|specialized|general purpose
Running (JUST GUESSING): D-Link embedded (95%), HP embedded (87%), Linux 5.X (87%), ETH Zurich Bluebottle (86%)
OS CPE: cpe:/h:dlink:di-524 cpe:/h:hp:m281fdn cpe:/o:linux:linux_kernel:5.11 cpe:/o:ethzurich:bluebottle
Aggressive OS guesses: D-Link DI-524 wireless broadband router (95%), HP M281fdn printer (87%), Proxmox Virtual Environment (Linux 5.11) (87%), Bluebottle OS (86%), HP LaserJet 2200dtn printer (86%), HP LaserJet 5M printer (86%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 28.98 seconds
```

Figure 20: Nmap OS detection scan on 192.168.1.0 identifying the device type as a WAP/printer with possible operating systems including D-Link embedded, HP embedded, and Linux 5.X.

```
C:\Users\user>nmap --script smb-enum-shares 192.168.1.0
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-04 00:37 +0300
Nmap scan report for 192.168.1.0
Host is up (0.024s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE    SERVICE
53/tcp    filtered domain
10778/tcp filtered unknown

Nmap done: 1 IP address (1 host up) scanned in 26.16 seconds
```

Figure 21: Nmap SMB enumeration scan on 192.168.1.0 showing two filtered ports (53/TCP - DNS and 10778/TCP - unknown) with no shared folders accessible, indicating restricted netwo